# *SleepVibe*: Sleep Stage Monitoring Through Active Bed-Frame Vibration

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Daily sleep stage monitoring is essential for assessing sleep quality and diagnosing sleep-related disorders. Recent approaches have explored scalable and affordable solutions for home sleep monitoring (e.g., wearables, cameras, RF-based sensing, passive vibration sensing). However, these approaches raise significant privacy and discomfort concerns and mostly suffer from inaccuracy due to the limited information that can be collected from users in different postures or under coverings.

In this paper, we present *SleepVibe*, the first sleep stage monitoring system through active bed-frame vibration sensing. The key insight is that sleep-related physiological processes (heartbeat, respiration, muscle twitches, etc.) alter the physical vibration response of the bed frame. However, these physiological effects are mixed together in the structural vibration response and do not consistently appear as distinct, measurable vibration signals. *SleepVibe* approaches this challenge by actively exciting the bed frame with an imperceptible interrogation vibration signal. This excitation, guided by structural characterization, enables these mixed vibration responses to be spectrally separable. Consequently, we identify and extract explainable sleep-related features beyond rate-level estimations from the separated vibration signal for sleep stage classification across various postures. We validate *SleepVibe* against an mmWave baseline in the real-world full- and partial-night sleep studies involving 34 young adults with no known severe sleep-related disorders (some reported mild obstructive sleep apnea). A consumer-grade EEG headband is used to provide reference sleep-stage labels. *SleepVibe* achieves up to 78% and 70% balanced accuracy in epoch-level KFold validation and leave-one-subject-out evaluation under full-night studies, exceeding mmWave by 11 and 6.4 percentage points, respectively. In addition, *SleepVibe* achieves 3.3 beats per minute (bpm) mean absolute error (MAE) for heart rate and 3.0 cycles per minute (cpm) MAE for respiration, outperforming state-of-the-art contact-free methods on heart rate by 4x while maintaining comparable respiratory accuracy.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computing methodologies** → **Machine learning**; • **Computer systems organization** → **Sensors and actuators**; **Embedded systems**.

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## 1 Introduction

Daily sleep-stage monitoring can reveal disruptions associated with sleep apnea, insomnia, narcolepsy, and rapid eye movement (REM) behavior disorder [3, 10, 66]. Healthy sleep typically progresses through wakefulness, light sleep, deep sleep, and REM, which support complementary aspects of physical and mental restoration [11, 55]. Yet roughly 56% of Americans experience irregular or fragmented sleep stages [66], motivating practical monitoring beyond the sleep laboratory.

Existing approaches to home sleep monitoring trade off comfort, privacy, and sensing reliability. Wearables embedded in headphones or watches infer sleep stages from heart rate, respiration, and related events, but require continuous skin contact and can cause discomfort or allergic reactions [6, 18, 24, 32, 37, 63]. Contact-free cameras, microphones, mmWave radar, and WiFi avoid body-worn sensors, yet blanket coverage and sleeping posture can reduce signal quality, while cameras and microphones raise bedroom privacy concerns [74, 77, 83, 86]. Piezoelectric films sense mattress deformation but require mattress modification and can be sensitive to wear and temperature [38, 56]. Structural vibration sensing avoids these constraints by recovering vital activity and sleep posture from vibrations transmitted through the bed [13, 16, 35, 43, 54, 69, 82]. However, passive vibration sensing captures weak, posture-dependent cardiac signals and provides limited characterization of respiration, making end-to-end sleep-stage monitoring difficult.

We present *SleepVibe*, an active bed-frame vibration-based sleep stage monitoring system. To the best of our knowledge, *SleepVibe* is the first system that leverages active sensing for human sleep monitoring. The key insight behind *SleepVibe* is that humans have a variety of sleep-related physiological processes that couple with the bed structure in distinct ways, thereby affecting the bed's structural vibration response. By injecting a well-designed and imperceptible vibration signal, we can extract and separate these coupled physiological processes.

We divide these sleep-related physiological processes into three different types: (1) Large sleep movements excite broader and stronger structural vibration responses (e.g., turning over in bed or muscle twitches). (2) Micro-vibrations of cardiac contractions that propagate through the mattress and bed frame via indirect body–structure coupling [43, 54, 82]. (3) Thoracic motion that creates slow, cyclic shifts in body load distribution imposed on the bed frame (e.g., respiration) [9]. Together, these heterogeneous processes are mixed into the physical vibration response of the bed frame (Figure 1), where differences in scale, frequency, and temporal dynamics coexist.

*SleepVibe* requires addressing a fundamental challenge: although these sleep-related physiological processes are mechanically present in the structural vibration response, they are highly mixed and only partially passively observable. Heartbeat-induced vibrations are relatively more apparent in the vibration response [13, 35, 43, 54, 82], but slower dynamics such as respiratory effort are largely underrepresented due to their sub-0.5 Hz nature [34, 70]. However, directly reducing these signals to rate-level estimates discards valuable sleep-related signal patterns contributing to robust sleep stage classification.

*SleepVibe* addresses this challenge by introducing active vibration sensing to separate coexisting physiological processes embedded in the structural vibration response. By injecting imperceptible excitation guided by structural characterization, active probing converts slow respiration-induced load redistribution into measurable vibration modulations while preserving heartbeat-induced components, enabling clean separation of the coupled responses under various postures.

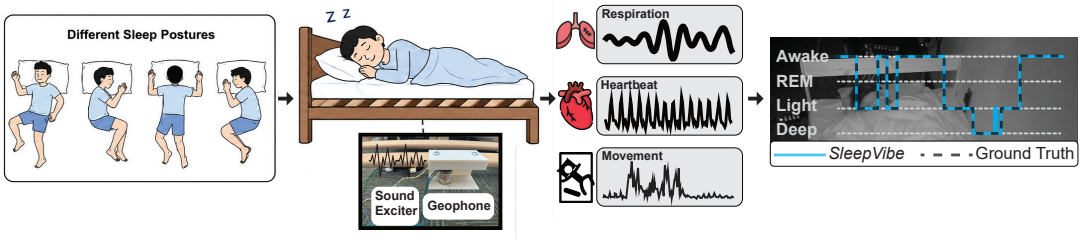



Fig. 1. *SleepVibe* achieves sleep stage monitoring by actively probing the bed structure and characterizing heartbeat activity, respiration, and body movement, which is robust across diverse sleep postures.

To ensure an effective and unobtrusive excitation, *SleepVibe* performs a one-time characterization to guide the excitation frequency selection. We first characterize the spectral components associated with heartbeat activity to identify the frequency bands to preserve for passive heartbeat sensing. We then characterize the resonant frequency bands of the bed structure, where small load variations produce pronounced changes in vibration response [85]. Based on these complementary characterizations, we select an excitation frequency at a structural resonance that remains spectrally distinct from the heartbeat-dominant components while compatible with imperceptible operation. The excitation amplitude is then calibrated below the sleeper's perceptual threshold. To further enhance signal fidelity, a mechanical amplification structure is incorporated to improve the detectability of weak heartbeat-induced vibrations.

Building on the separated vibration waveforms, *SleepVibe* identifies and models a physiology-aware feature set that is explainable within the vibration modality. Rather than relying solely on rate estimation, we systematically derive interpretable statistical and physiological features that characterize how cardiac activity, respiration, and movement manifest in the structural vibration response. These features capture waveform morphology, temporal variability, and cross-signal relationships that are observable within the vibration modality and physiologically associated with sleep-stage transitions, thereby establishing an interpretable connection between sleep physiology and the measured structural response. In the end, *SleepVibe* employs a bidirectional gated recurrent unit (BiGRU) model [14] to classify sleep stages from these physiology-aware feature representations with additional temporal context across consecutive sleep epochs.

In summary, our work makes the following contributions:

(1) We present *SleepVibe*, the first sleep stage monitoring system based on active bed-frame vibration sensing. By actively probing the structure, *SleepVibe* separates sleep-related physiological processes, enabling reliable recovery of heartbeat, respiration, and movement-related metrics from a single vibration modality.

(2) We model a set of sleep-relevant physiological features represented through the structure-borne vibration. Enabled by active probing, these features capture explainable information such as waveform morphology beyond rate-level metrics, supporting interpretable and robust sleep stage classification.

(3) We evaluate *SleepVibe* across 34 subjects without known severe sleep-related disorders aged 20 to 28 years in real-world full- and partial-night sleep (some reported mild obstructive sleep apnea). Using a commercial sleep-monitoring headband [7, 42] as reference, *SleepVibe* achieves up to 77.6% and 69.9% balanced accuracy under KFold validation and leave-one-subject-out (LOSO) evaluation, exceeding the mmWave baseline by 10.9 and 6.4 percentage points. In addition, *SleepVibe* estimates heart rate with an MAE of 3.3 bpm, approximately a 4x improvement over mmWave estimations, and respiratory rate with an MAE of 2.97 cpm, falling within commonly reported tolerance ranges.

(4) We further evaluate the robustness of reconstructed heartbeat and respiratory signals under various physiological and structural variations, including various postures, elevated heart and respiratory rates,

different floor materials, wooden bed frames, and varying mattress thicknesses up to 14 inches. These evaluations demonstrate the robustness of active bed-frame vibration sensing across a range of realistic deployment conditions.

## 2 Related Work

### 2.1 Sleep Stage Modeling

Understanding the structures of sleep stages is crucial for evaluating sleep quality, which plays a fundamental role in physical health, cognitive performance, and emotional well-being [3, 10, 66]. Different sleep stages serve distinct physiological and neurological functions, and imbalances among them can reflect or contribute to health issues. Beyond its value in clinical diagnosis, tracking sleep stages also benefits everyday life. Daily monitoring at home allows individuals to gain insight into their sleep habits, recognize patterns or disturbances, and improve overall well-being [41, 60].

Sleep stages are distinguished by systematic changes in movement, cardiac activity, and respiration. Wakefulness generally involves more frequent movement and variable cardiac and respiratory patterns. Non-REM sleep exhibits progressively reduced movement, slower cardiac activity, and more regular respiration, and REM sleep combines muscle atonia with irregular autonomic activity [55, 66]. Consumer sleep monitors group N1 and N2 commonly as light sleep and label N3 as deep sleep [58]. These stage-dependent physiological signatures motivate sleep-monitoring systems that infer sleep stages from movement, heartbeat, and respiratory signals.

### 2.2 Sensing Technologies for Sleep Monitoring

Polysomnography (PSG) is the clinical gold standard for sleep staging, combining EEG, EOG, ECG, and electromyography (EMG) for expert interpretation [22, 61, 65]. However, its cost, labor-intensive setup, and body-attached electrodes limit routine home use and can disrupt natural sleep [30]. Many prior works have explored sleep stage monitoring by combining heart rate, respiratory rate, and sleep movement, which aligns well with home settings due to the reduced cost of required sensors and lower deployment burden. Wearables such as Fitbit [18] and Apple Watch [64] combine photoplethysmography (PPG) with accelerometers [18, 59, 64, 76] for sleep stage monitoring. Similar sensing has been integrated into in-ear devices [32]. Although convenient, prolonged body contact can cause discomfort or skin irritation [6, 37]. Contact-free systems instead use amplified video motion, sleep-related sounds, mmWave, or WiFi channel information [74, 82, 83, 86]. These systems reduce on-body burden, but cameras and microphones raise privacy concerns, while occlusion, posture, and environmental conditions can reduce the sensing robustness of mmWave and WiFi. Mattress-integrated piezoelectric films provide another unobtrusive option by measuring pressure and vibrations associated with cardiac activity, respiration, and movement [23, 38, 56, 67]. Yet, it suffers from sensor wear, temperature sensitivity, and static loading, limiting long-term reliability.

### 2.3 Structural Vibration Sensing

Structural vibration sensing has its origins in civil and mechanical engineering, where it is used to monitor the health of large-scale infrastructures and properties of vibration sources. For example, Liu et al. [44] and Ntotsios et al. [51] characterize the bridge health based on the vibration induced by drive-by vehicles. Beyond structural diagnostics, vibrations also reveal information about the source itself: Richardson et al. [62] and O'Brien et al. [52] estimate vehicle weight under usual driving conditions by sensing vehicle-induced vibrations on the road. Wang et al. [75] introduces how machine-induced vibrations can serve as indicators of health for industrial equipment.

This principle has inspired the development of human-centered sensing systems that treat the body as a dynamic source of mechanical excitation. For example, Pan et al. [53] leverages the geophone sensor to capture structural vibrations generated by footsteps for user identification and localization. Dong et al. [21] further

Table 1. Comparison of representative sleep monitoring systems and sensing modalities. The table highlights sensing capabilities relevant to fine-grained sleep stage classification.

| System | Contact-free | Occlusion-tolerant | Posture-tolerant | Privacy-friendly | Long-term Usage | Heartbeat Waveform | Resp. Effort | Movement | Sleep Stages |
|---|---|---|---|---|---|---|---|---|---|
| PSG (EEG/EOG/ECG/EMG) [22] | ✗ | ✓ | ✓ | ✓ | ✗ | ✓ | ✓ | ✓ | ✓ |
| EarSleep [32] | ✗ | ✓ | ✓ | ✓ | ◯ | ✓ | ✓ | ✓ | ✓ |
| Camera [74] | ✓ | ✗ | ✗ | ✗ | ✓ | ✗ | ✓ | ✓ | ✓ |
| Microphone [86] | ✓ | ✓ | ✓ | ✗ | ✓ | ✗ | ✓ | ✓ | ✓ |
| mmVital [40] | ✓ | ✗ | ✗ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Piezoelectric Mattress Sensors [38, 56] | ✓ | ✓ | ✓ | ✓ | ✗ | ✓ | ✓ | ✓ | ✗ |
| Passive Vibration Sensing [35, 43, 69] | ✓ | ✓ | ◯ | ✓ | ✓ | ✓ | ◯ | ✓ | ✗ |
| HeartQuake [54] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✗ | ✓ | ✗ |
| ***SleepVibe*** | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

**Legend:** ✓ Supported; ◯ Partially supported / limited fidelity; ✗ Not supported.

deepens the analysis and adopts structural vibration sensing to monitor gait health. These works demonstrate the feasibility of capturing body movement through structural coupling.

## 2.4 Sleep Monitoring Through Passive Vibration Sensing

Passive vibration sensing has been explored for heart/respiration rate estimation [13, 35, 43, 54] and in-bed posture classification [69, 81] through geophone-based vibration sensors [70]. These systems capture heartbeat-induced vibrations transmitted through the body and surrounding structures [13, 16, 35, 43, 54]. Some further infer respiration from heartbeat amplitude modulation [35, 43], and leverage posture-dependent vibration patterns to support sleeping posture classification [69]. However, prior work generally treats vital-rate estimation and posture recognition as separate tasks, leaving reliability across sleep postures unclear. Posture changes the body–bed coupling path, reshaping vibration amplitude and morphology and potentially confounding respiration inferred from heartbeat modulation. Moreover, rate-level outputs discard temporal dynamics, waveform morphology, and interactions with movement. These limitations restrict the use of passive vibration sensing for robust and interpretable sleep-stage inference.

To address these limitations, *SleepVibe* integrates active bed-frame vibration sensing into an end-to-end sleep-stage monitoring system. Its imperceptible active excitation provides a controllable structural response that helps separate heartbeat, respiration, and movement components across sleep postures. From the recovered waveforms, *SleepVibe* derives explainable features describing morphology, temporal variability, and cross-process relationships beyond rate-level representations, enabling more robust and interpretable sleep-stage inference.

# 3 Design of *SleepVibe*

Guided by the structural characterization of the bed frame, *SleepVibe* exploits an active vibration sensing framework to extract sub-0.5 Hz respiratory effort, weak heartbeat-induced vibrations, and contextual body movement cues from mixed vibration responses (Figure 2). *SleepVibe* begins with vibration sensing instrumentation, integrating geophone sensors, controlled mechanical excitation, and physical amplification structures. The mixed vibration signals are then processed and separated in the frequency domain, isolating the heartbeat and respiratory effort components. Finally, statistical and physiological features extracted from these signals, together with detected movement cues, are integrated into a tailored machine learning model to enable sleep stage classification.

## 3.1 Vibration Sensing Instrumentation

*3.1.1 Respiration.* Respiration occurs primarily below 0.5 Hz [9, 15], outside the effective bandwidth of geophones. At these near-DC frequencies, breathing appears mainly as a quasi-static redistribution of body load rather than

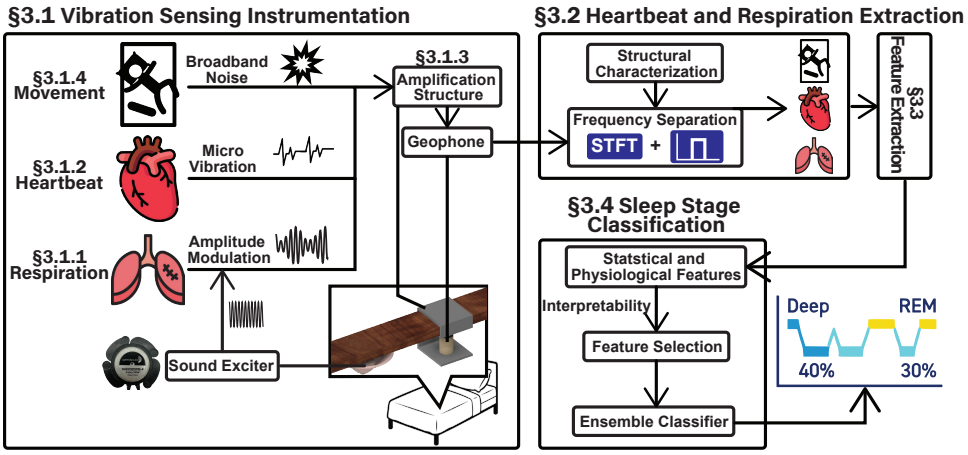



Fig. 2. System Overview of *SleepVibe*

as propagating vibration, making it difficult to distinguish from drift and environmental noise using passive bed-frame measurements [34, 70].

To address this limitation, we activate the bed frame with an imperceptible vibration as a carrier through an attached sound exciter [17]. The key idea is that the associated thoracic motion induces slight pressure changes on the bed frame, altering the structural vibration response and consequently modulating the amplitude of the excitation signal. It enables geophones to reliably characterize the full respiratory process by capturing the changing vibration responses. To maximize excitation's sensitivity to pressure changes, we leverage an impulse signal to characterize the resonant frequency bands of the instrumented beams (Figure 9b). Based on this characterization, we select an excitation frequency at the dominant band that maintains spectral separation from heartbeat-induced vibrations, while supporting imperceptible operation. The excitation amplitude is further tuned to remain imperceptible to the sleeper.

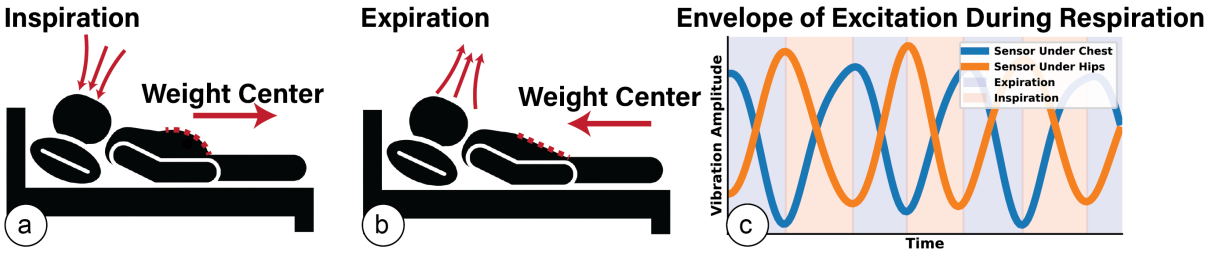



Fig. 3. The insight of the active vibration sensing for respiration. (a) During inspiration, the body weight center shifts toward the leg of the bed. (b) During expiration, the body weight shifts back. (c) This diaphragmatic breathing leads to a slow-cycle body weight shift that impacts the excitation signal amplitude following the respiration cycle.

This active vibration sensing framework exploits respiration-induced shifts in body weight distribution, which persist across sleep postures [9, 36]. As shown in Figure 3a&b, diaphragmatic contraction during inspiration shifts the body's center of mass slightly toward the foot of the bed, while expiration shifts it back toward the head. These cyclic load changes modulate the bed's resonant response and, consequently, the vibration amplitude measured by the geophones [80]. As illustrated in Figure 3c, respiration therefore appears as a low-frequency amplitude

modulation of the injected carrier, providing a posture-robust proxy for respiratory effort. We quantitatively validate the relationship between applied load and excitation amplitude in Section 4.2.

*3.1.2 Heartbeat Activities.* Reliable heartbeat characterization under active vibration sensing requires preserving the weak heartbeat-induced vibration response while probing the bed structure for respiration. Heartbeat generates impulse-like micro-vibrations that are mechanically embedded and observable in the vibration response. However, these signals are easily masked by movement, and the intrinsic signal dynamics beyond rate-level information might be corrupted by direct excitation. Thus, the active excitation requires careful consideration to preserve the original heartbeat-induced vibration responses.

To enable a clean separation between heartbeat and respiration, we characterize the bed's structural vibration response to identify the dominant frequency components of heartbeat. Experiments with human subjects show that heartbeat consistently excites weak but distinct vibration components concentrated around 10 Hz, as illustrated in Figure 9a, which coincides with the natural frequency of the employed geophone [70]. Importantly, this band is spectrally distinguishable from the selected excitation frequency for active probing. We further quantify the interaction between the heartbeat-dominant band and the excitation component in Section 4.2, demonstrating that the selected frequency allocation preserves heartbeat characterization while supporting excitation-based respiratory sensing.

*3.1.3 Mechanical Amplification of Vibration.* We incorporate a mechanical amplification structure to improve signal fidelity under imperceptible active excitation and enhance the detectability of weak heartbeat-induced vibrations. As shown in Figure 2, the design consists of a C-shaped frame with a suspended plate, where the thin vertical wall acts as a compliant element connecting the plate to the bed frame. This configuration converts small base vibrations near resonance into larger displacements at the plate, while the lever-arm effect concentrates vibration at the geophone-mounted free edge. As a result, heartbeat-induced micro-vibrations are mechanically amplified, increasing the effective signal margin and reducing posture-dependent attenuation.

*3.1.4 Vibration Sensing for Sleep Movements.* In addition to heartbeat and respiration, body movements during sleep couple strongly with the bed structure and generate vibration signals with substantially higher amplitude and broader spectral content. These movements range from large posture adjustments, such as turning over, to transient motions including limb or muscle twitches. Characterizing the timing and extent of such movements is important for sleep monitoring, as movement is closely associated with sleep stage transitions [57] and periods of unreliable vital signal estimation.

## 3.2 Heartbeat Activity, Respiration, and Sleep Movement Extraction

Guided by the bed-frame characterization, we separate respiratory and heartbeat responses according to their distinct frequency signatures, as shown in Figure 4. For each bed frame, we perform a one-time characterization to identify a broad excitation band that is spectrally separated from heartbeat-dominant components. The excitation amplitude is then calibrated for each participant to remain below the individual sleeper's perceptual threshold.

The separated signals are then passed through the dedicated band-pass filters shown in Figure 4 to isolate their physiological components. We extract respiration with a 0.1–0.4 Hz band-pass filter, which covers the normal respiratory rate range of a resting adult and child [15, 25]. Heartbeat is extracted with a 0.7–3.0 Hz filter covering 42–180 bpm, which encompasses the lower rates observed during sleep (down to 40 bpm), the normal resting range of 60–100 bpm [8], and higher rates occasionally seen in athletic individuals [33]. For each analysis window, we model a set of sleep-related physiological features as discussed later in Section 3.3.

We employ a threshold-based movement classifier to identify noise-dominant periods caused by large body motions. For each of the two geophone heartbeat-envelope channels, we compute the signal variance over non-overlapping one-second windows and derive a channel-specific threshold from that night's variance distribution.

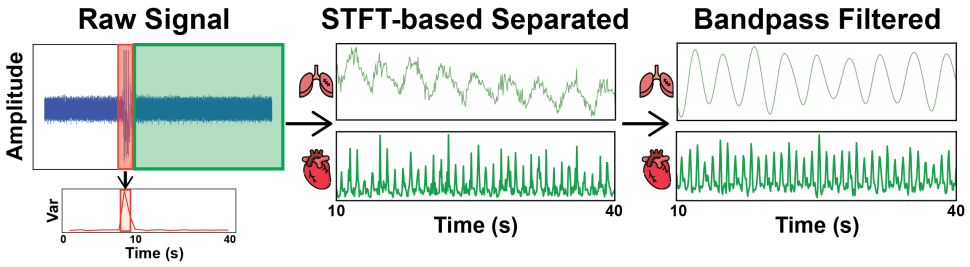



Fig. 4. *SleepVibe* signal processing pipeline.

The threshold is defined as $Q_3 + m \times \text{IQR}$, where $Q_3$ and IQR denote the third quartile and interquartile range, respectively. The multiplier ($m$) is calibrated based on the sensor agreement of the extracted motion masks. The resulting binary motion trace is summarized within each 30s epoch into different features retained as the input of the sleep-stage classifier.

## 3.3 Physiological Features Characterization

Table 2. Summary of physiological features used for sleep-stage classification (83 total)

| Category | Features | Physiology | Count |
|---|---|---|---|
| Basic Statistics | Mean, standard deviation, minimum, maximum, median | Vital signal amplitude and variability | 25 |
| Cross-Channel Correlation | Heart–heart, respiration–respiration, cross-modal correlations | Signal consistency and heart-breath coupling | 3 |
| Waveform Frequency Domain | Dominant frequency, low-band power (0.1–1 Hz), high-band power (1–5 Hz) | Spectral content of each 30s vibration waveform | 14 |
| Peak/Valley Detection | Peak/Valley count, Peak/Valley amplitude, Peak Valley range, Peak regularity | Cardiac and respiratory rhythm characteristics | 30 |
| HRV-Related | *HRV*, LF power (0.04–0.15 Hz), HF power (0.15–0.40 Hz), LF/HF ratio, total power (0.0033–0.40 Hz), normalized LF/HF power | Time/Spectral features of beat-to-beat intervals | 7 |
| Others [27] | $P_{sdm}$ (Inhalation depth), $T_{sdm}$ (Exhalation depth), *DA* (Combination of changes in movement and heart rate), *BM* (Body Movement) | Heartbeat, respiratory volume, and movement magnitude metrics | 4 |
| **Total** | | | **83** |

We characterize 83 physiological and explainable features for sleep stage monitoring based on the extracted structural vibration responses. Guided by prior work on sleep stage inference from cardiorespiratory dynamics [27, 78], we curate a vibration-specific feature set that captures multiple complementary aspects of the actively probed structural response [79]. As summarized in Table 2, these features include basic statistical indicators, frequency-domain characteristics, peak and valley dynamics, cross-channel correlations, and other physiologically motivated measures such as heart rate variability (HRV). Collectively, these features encode waveform morphology, temporal variability, and cross-signal relationships that are known to correlate with sleep stages. Figure 5 illustrates representative examples of how these features are expressed in the vibration waveform, highlighting how sleep-related physiology is reflected in the actively sensed structural response.

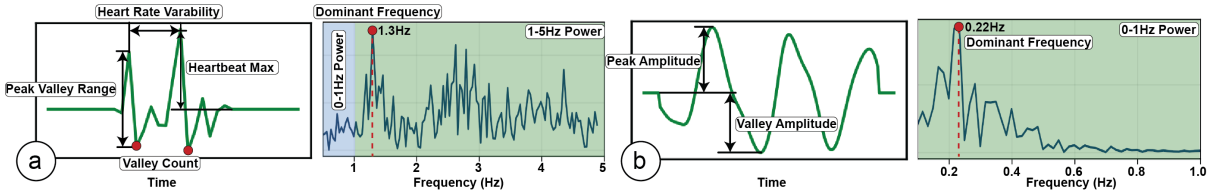



Fig. 5. Examples of (a) heartbeat-related features and (b) respiration-related features, which are extracted from separated vibration responses.

## 3.4 Sleep Stage Classification

To capture temporal dependencies across consecutive 30s epochs, we use a compact MLP–BiGRU classifier [14]. As shown in Figure 6, the MLP maps physiological features of each epoch to a latent representation, which is concatenated with a fixed Fourier encoding of normalized time-of-night [72]. The BiGRU processes the resulting sequence in both temporal directions, allowing each prediction to use both physiological information and surrounding sleep context.

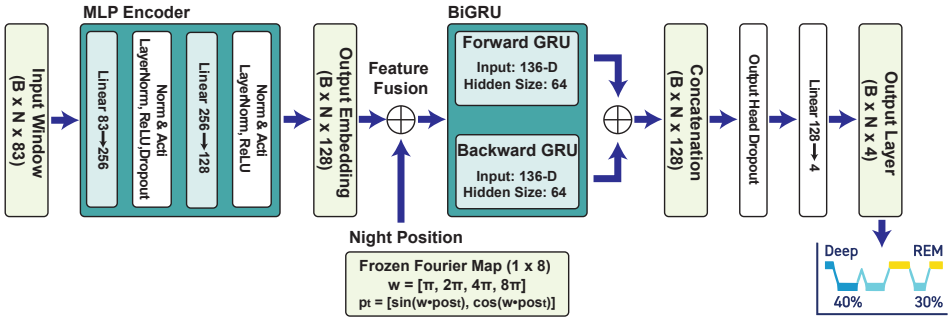



Fig. 6. The MLP-BiGRU model architecture.

The training objective reduces bias toward common Light sleep stages by combining inverse-frequency class weighting with focal loss [12]. For class $c$, we use $w_c = \left[N/C(n_c+1)\right]^{\rho}$, where $n_c$ and $N$ are the numbers of class-$c$ and total training epochs, $C = 4$, and $\rho$ controls the reweighting strength. The additive constant prevents division by zero. The resulting loss is $\mathcal{L} = -N^{-1}\sum_{i=1}^{N} w_{y_i}(1-p_{i,y_i})^{\gamma}\log p_{i,y_i}$, where $y_i$ is the ground-truth class, $p_{i,y_i}$ is its predicted probability, and $\gamma$ is the focusing parameter. Class weighting increases the contribution of underrepresented stages, while focal loss emphasizes difficult epochs.

Finally, the model operates on sequences of consecutive 30s epochs. Each window is represented by 83 standardized physiological features, yielding an input tensor of size $N \times 83$. The MLP maps each epoch to a 128-dimensional embedding, which is concatenated with an eight-dimensional Fourier time encoding before entering the BiGRU. Thus, the BiGRU input has size $N \times 136$. The output has size $N \times 4$, providing one prediction over the four stages: Awake, REM, Light, and Deep, for each epoch. Feature normalization is computed from the training data and then applied unchanged to the validation and test data.

# 4 Experimental Setup

## 4.1 Geophone Sensor Unit and Bed Instrumentation

We employ the vibration sensing module developed in [16] and the sound exciter to probe the bed frame's vibration response. The sensing module is composed of: (1) A SM-24 geophone [70], (2) an amplifier & filter circuit [16], and (3) an analog-to-digital converter (ADC) from Analog Discovery [19], as shown in Figure 7a, b, and c. The SM-24 geophone focuses on the lower frequency range between 10–240 Hz, making it well-suited for bed structural vibration measurements reported in prior studies [16, 84]. All geophones are held by the C-shaped frame that amplifies the vibration response from physiological processes (Figure 7d). The sound exciters (Figure 7e) are rigidly attached to the same bed-frame beam as the geophones. To enhance the quality of the recorded signals and suppress electrical interference, we employ a sensing board featuring a programmable non-inverting amplifier combined with a second-order Sallen–Key low-pass filter[16]. Finally, we use the Analog Discovery Pro 3000 (ADP3000) to collect data synchronously.




Fig. 7. A vibration sensing module is composed of (a) a geophone and (b) an amplifier&filter board, connected to one channel of (c) ADP3000. Each geophone is mounted on our (d) amplification structure. (e) Sound exciter for active vibration sensing.

To determine sensor placement, we use a pressure mat [45] to characterize respiration-induced pressure changes under supine and left-lateral postures. As shown in Figure 8b, pressure varies most consistently beneath the chest and hip during inspiration and expiration, identifying these regions as dominant load-bearing locations. The chest location also provides favorable coupling for heartbeat-induced vibrations due to its proximity to the heart. Accordingly, we deploy two geophone pairs beneath the chest and hip regions, as shown in Figure 8a, and mount two sound exciters at separate bed frame locations for active structural probing.

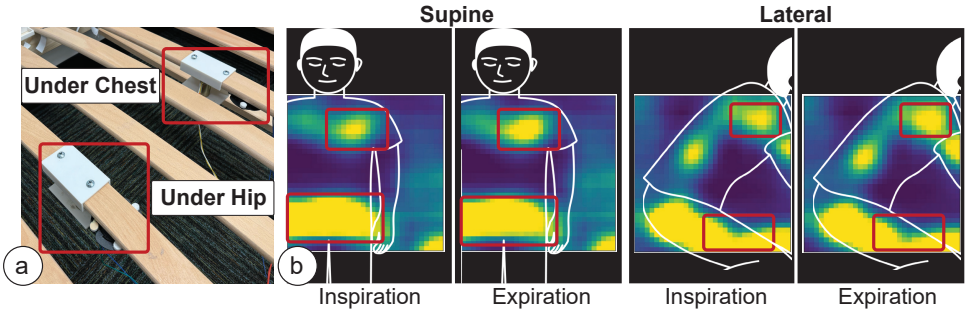



Fig. 8. (a) The optimal geophone placement under the chest and hip regions is guided by (b) the characterization of pressure distribution during respiration.

## 4.2 Structural Characterization

*Deployment and Calibration Protocol.* We characterize the bed frame's structural vibration response to guide excitation signal design, enabling spectral separation between heartbeat-induced vibrations and respiration-modulated active excitation while maintaining sleeper comfort. Figure 9 presents the spectrum of heartbeat-induced vibrations and the frequency-domain impulse response of the adopted bed structure. As shown in

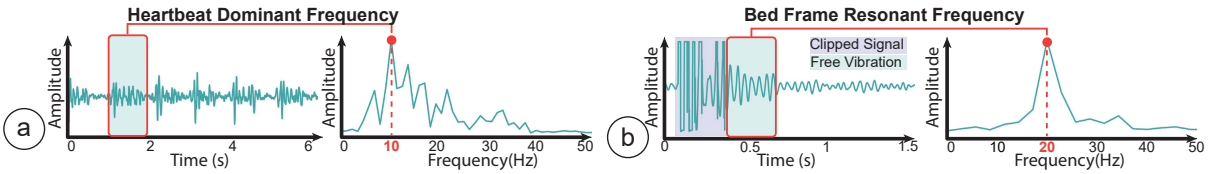



Fig. 9. (a) The dominant frequency of heartbeat-induced vibration is around 10Hz, and (b) the peak resonant frequency of the bed frame is around 20Hz.

Figure 9a, heartbeat-induced vibrations exhibit a dominant component around 10 Hz, which is consistently observed across our pilot and sleep studies and is likely influenced by the natural frequency of the adopted geophone. We therefore treat this frequency range as the heartbeat-dominant band when selecting the active excitation frequency.

For structural characterization, we use the installed geophone sensing modules, acquisition system, and sound exciters described in Section 4.1. With the mattress removed, we apply a series of hammer impacts to the sensor-mounted beam over 30 seconds and record the resulting free-vibration responses. For each deployment, we select a prominent structural resonance above the heartbeat-dominant band around 10 Hz, prioritizing a resonance with the highest spectral peak and sufficient separation from the cardiac component. Figure 9b shows an example characterization, in which a prominent resonance is observed around 20 Hz. Excitation at the selected resonance provides a strong structural response, while the excitation amplitude is subsequently calibrated to remain below the participant's perception threshold. We repeat this structural characterization when using a different bed frame or after reassembling the frame, because these operations might change the resonant vibration responses, as shown later in Section 5.6.6.

We calibrate the excitation amplitude for each participant by gradually increasing it until the participant first perceives the vibration and then reducing it until it becomes imperceptible across sleep postures. We verify the setting by switching the excitation on and off at least three times and accept it only when the participant consistently reports no perceptible difference. This participant-specific amplitude calibration takes no more than three minutes.

*Validation of Frequency-Domain Separation.* To verify the spectral isolation of the selected frequencies for heartbeat and respiratory monitoring, we validate whether the heartbeat-induced response and active excitation can be independently recovered from the same geophone measurement. As shown in Figure 10a, the raw signal contains the heartbeat-induced vibration, where each pulse generates a 10 Hz-dominated peak (Figure 9) and the active excitation carrier at 20 Hz. Suppressing the 20 Hz band preserves a clear heartbeat signal aligning with the PPG-extracted ground truth. Conversely, suppressing the 10 Hz band preserves the 20 Hz carrier and its respiration-modulated amplitude envelope. The resulting envelope remains aligned with the synchronized respiration reference. These complementary filtering results demonstrate sufficient spectral separation for independently extracting heartbeat and respiration information from the same vibration signal.

*Validation of Load-Induced Amplitude Modulation.* To quantitatively analyze the effect of respiration-induced body weight shifts, we conduct a controlled loading experiment shown in Figure 10b. We place a bottle directly above the instrumented bed beam and progressively increase its mass from 1.3 kg to 3.0 kg by adding salt, while holding the excitation frequency and drive amplitude constant. Across 18 loading conditions, the measured carrier amplitude decreases systematically with increasing mass and exhibits a strong negative linear relationship with the applied load ($R^2 = 0.954$, Pearson's $r = -0.977$, $p < 0.001$). This result is consistent with the proposed mechanism, in which cyclic respiration-induced load redistribution modulates the amplitude of the active excitation signal.

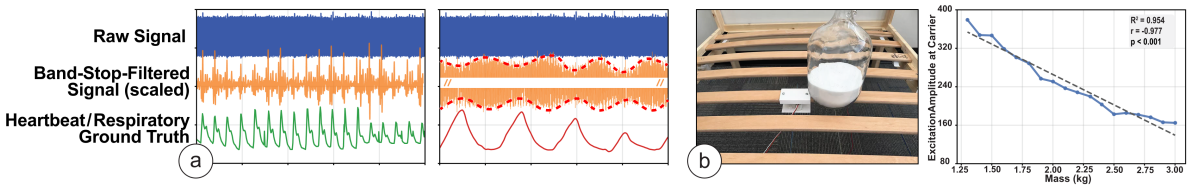



Fig. 10. Validation of frequency-domain separation and load-induced amplitude modulation. (a) Raw geophone signals and the signals after band-stop filtering for heartbeat and respiration. (b) Controlled loading experiment using a salt-filled bottle placed above the instrumented bed beam.

Appendix B further shows the sensor bracket amplification effect: the C-shaped structure-mounted sensor exhibits higher vibration amplitudes than the bed-mounted one across a broad frequency range.

## 4.3 Sleep Study

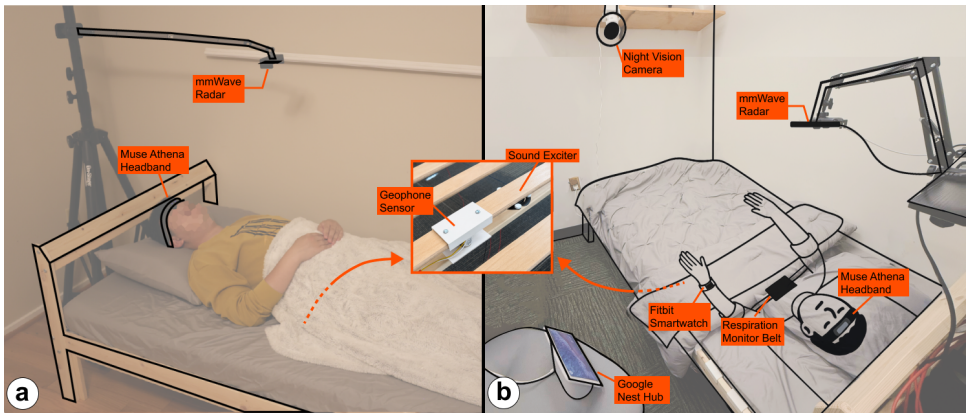



Fig. 11. The data collection environments: (a) full-night sleep recording and (b) partial-night sleep recording.

We evaluate *SleepVibe* through two complementary sleep studies. We first conduct a full-night study in which eight participants each complete three overnight sessions, enabling us to assess sleep-stage classification under natural, long-duration sleep and repeated use (as shown in Figure 11a). We then conduct a partial-night study with a larger cohort of 26 participants, each completing a 3-hour sleep session, to broaden the evaluation across individuals and compare *SleepVibe* against representative off-the-shelf systems (as shown in Figure 11b). Together, these studies provide complementary evidence of system performance across complete overnight recordings, repeated measurements, and a more diverse participant cohort. Before each study, we calibrated the excitation amplitude separately for each participant using the perceptibility checks described in Section 4.2. All data collection procedures were approved by the Institutional Review Board (IRB), and all collected data were de-identified and securely stored in accordance with the approved protocol.

To minimize participant burden while obtaining physiological reference measurements, we selected minimally intrusive sensing devices. For sleep-stage annotation, rather than using a full PSG setup, we employed a Muse headband equipped with EEG and functional near-infrared spectroscopy (fNIRS) sensing [4, 7, 42]. The device provides sleep-stage labels at 30s intervals. For the mmWave baseline, we adopted the algorithm proposed in [40] to extract respiratory and heartbeat waveforms from the radar signal at a sampling rate of 20 Hz.

# 5 Evaluation

## 5.1 Full-Night Study

The full-night study evaluates *SleepVibe* under repeated, natural overnight use. We recruited eight adults aged 20–28 years with no known severe sleep-related disorders (four males and four females; some reported mild obstructive sleep apnea), and each completed three overnight sessions in a real bedroom with a 4-inch mattress (Figure 11a). Out of 24 recorded nights, five were excluded because of setup errors such as failure to initiate data collection, leaving 19 nights and approximately 136 hours and 40 minutes for analysis. We performed the structural characterization and calibration described in Section 4.2, identifying a 40 Hz resonance and calibrating the excitation amplitude to remain imperceptible to users. We used mmWave as the sleep-stage classification baseline.

The dataset contains 16,401 labeled 30s epochs for *SleepVibe* and 16,437 for mmWave, resulting in 7.41% Awake, 26.15% REM, 45.95% Light, and 20.48% Deep. The model uses the 83 features described in Table 2. We retain all epochs for sleep-stage classification, including motion-dominant epochs, and characterize their movement information using the extracted motion features. As Light sleep dominates the naturally imbalanced dataset, we mitigate class imbalance by combining inverse-frequency class weighting with focal loss. At the end of the study, each participant completed a post-sleep questionnaire to assess whether *SleepVibe* affected their sleep experience.

*5.1.1 Epoch-Level KFold Validation.* We first assess within-cohort performance using shuffled, stratified 10-fold cross-validation. For each modality, sequences from all participants are pooled and divided into eight training folds, one validation fold, and one test fold. Depending on the sequence length and fold assignment, the training partition contains 12,760–13,151 sleep epochs, while the validation and test partitions each contain 1,560–1,644 epochs. We train two separate models using the same 83-feature representation for *SleepVibe* and mmWave. Feature normalization is estimated exclusively from the training folds and then applied unchanged to the validation and test data.

Both models are optimized with AdamW [87] using a learning rate of $6 \times 10^{-3}$, weight decay of $10^{-4}$, batch size 64, and cosine learning-rate decay. The training objective combines focal loss $\gamma = 0.1$ with inverse-frequency class weights $w_c = [N/(4(n_c + 1))]^{0.05}$, where $N$ denotes the total number of 30s epochs in the training partition and $n_c$ denotes the number assigned to class $c$. We use validation balanced accuracy for early stopping, restore the best validation checkpoint before testing, and repeat the complete 10-fold evaluation with five random seeds.

*SleepVibe* consistently outperforms mmWave across all evaluated sequence lengths as shown in Figure 12a. At $L = 1$, *SleepVibe* achieves a balanced accuracy of 77.6% ± 0.1%, exceeding mmWave (66.7% ± 0.3%) by 10.9 percentage points. The advantage remains at $L = 40$, where *SleepVibe* achieves 66.7% ± 0.7% compared with 59.3% ± 0.7% for mmWave. Figure 12b illustrates a class-level view of this improvement at $L = 1$. *SleepVibe* yields higher recall for all four stages, reaching 79.8% for REM and 82.1% for Deep, which correspond to gains of 16.9 and 10.8 percentage points over mmWave, respectively.

The decline in KFold performance with increasing sequence length reflects a trade-off between temporal context and sequence availability. Increasing $L$ from 1 to 40 allows the model to observe a longer segment of sleep progression, but reduces the mean test-fold size from approximately 1,640 to 40 sequences. The number of evaluated 30s epochs remains around 1,600 because the model produces a prediction for every epoch within each sequence. Nevertheless, the smaller number of distinct sequences makes each fold more sensitive to its participant and sleep-stage composition. Accordingly, the fold-level standard deviation of macro-F1 increases from 1.0–1.1 to approximately 5.0–6.5 percentage points. The $L = 1$ setting therefore provides more training instances and more stable fold-level estimates, although adjacent epochs may occur in different folds and yield somewhat optimistic performance.

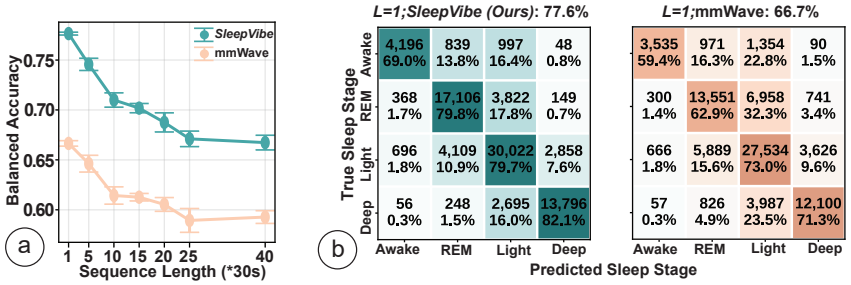



Fig. 12. Full-night KFold results. (a) Balanced accuracy across sequence lengths. (b) Confusion matrices at $L = 1$.

*5.1.2 Leave-One-Subject-Out Evaluation.* We next use leave-one-subject-out (LOSO) evaluation to assess whether *SleepVibe* generalizes to an unseen participant. In each of the eight folds, all nights from one participant are held out as the test set. From the other seven participants, we reserve one complete night for validation and use all remaining nights for training. The validation night is rotated across the five random seeds.

The model architecture and training recipe remain the same as in the KFold evaluation. Before sequence construction, we apply per-night feature normalization using unlabeled statistics from each recording [71]. Since participants are fully separated across splits, we form half-overlapping sequences in training, validation, and testing, and average the logits from all sequences containing the same 30s sleep epoch during inference before assigning its stage. Across eight held-out participants and five random seeds, this procedure produces 40 model fits for each sequence length and sensing modality.

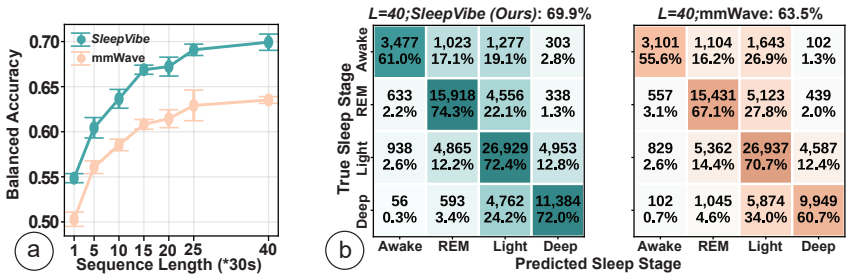



Fig. 13. Full-night LOSO results. (a) Participant-balanced accuracy across sequence lengths. (b) Confusion matrices at $L = 40$.

In contrast to the KFold results, LOSO performance improves as the temporal context increases, and *SleepVibe* remains more accurate than mmWave throughout the sequence-length sweep, as shown in Figure 13a. Each LOSO test fold contains all recordings from one participant and therefore preserves a larger, more continuous sleep progression trajectory than an individual KFold test partition. We first compute balanced accuracy separately for each held-out participant and then average the eight participant-level scores, ensuring that participants with longer recordings do not receive greater weight. The error bars show variation across the five random seeds. As $L$ increases from 1 to 40, balanced accuracy rises from 54.9% ± 0.5% to 69.9% ± 0.9% for *SleepVibe* and from 50.3% ± 0.8% to 63.5% ± 0.4% for mmWave. In addition, *SleepVibe* achieves a Cohen's $\kappa$ of 0.559, compared with 0.509 for mmWave, indicating stronger agreement with the consumer-grade EEG sleep-stage labels and reaching the moderate-agreement range.

Figure 13b further shows the confusion matrices at $L = 40$. *SleepVibe* improves recall across all four sleep stages. The largest improvement occurs for Deep sleep, where recall increases by 11.3 percentage points relative

to mmWave. Recall also improves by 7.2 percentage points for REM and 5.4 percentage points for Awake, while the two modalities perform similarly for Light sleep. Overall, these results suggest that *SleepVibe* generalizes more effectively to unseen participants, with a particularly strong advantage in recognizing deep sleep.

*5.1.3 Evaluation of Perceptibility and Sleep Disturbance.* To evaluate whether the active vibration affected participants' sleep experience, we conducted a post-study questionnaire using a five-point Likert scale [5]. The excitation itself was effectively imperceptible. All participants disagreed that they noticed sound or vibration from the system: 87.5% strongly disagreed and 12.5% disagreed. The same distribution was observed when participants were asked whether the excitation affected their sleep quality (Figure 14a). By comparison, 50% reported an effect from the wearable reference devices, 12.5% from the face-down LED on the mmWave device, and 50% from sleeping in an unfamiliar room (Figure 14b). When asked to identify specific sources of disturbance, all participants strongly disagreed that active vibration contributed, whereas 37.5% identified the EEG headband, 12.5% the mmWave device, and 37.5% the room environment (Figure 14c).

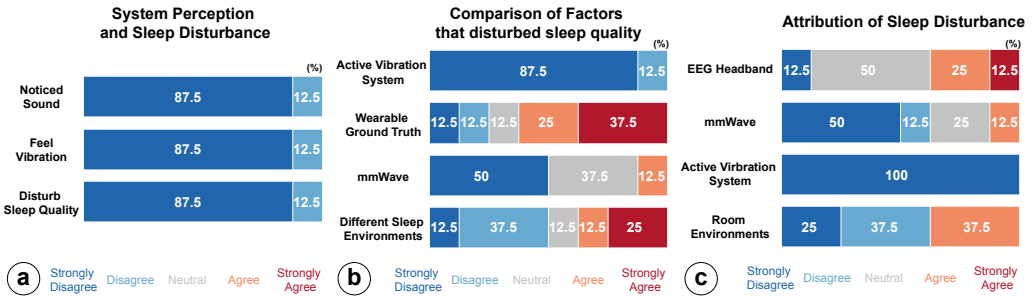



Fig. 14. Five-point Likert evaluation of (a) active-excitation perceptibility, (b) effects of experimental components on sleep quality, and (c) attributed sources of sleep disturbance.

In addition, we quantitatively measured the sound level at the pillow location using a sound-level meter. At the highest excitation amplitude calibrated across the eight participants, the measured sound level increased by only 0.6 dB, from 49.6 dB with the system off to 50.2 dB with the system on. This small increase provides additional objective evidence that the active excitation introduced minimal additional audible sound.

## 5.2 Partial-Night Study

The partial-night study complements the repeated full-night recordings with a larger cohort and direct comparisons against off-the-shelf sleep-monitoring systems. We recruited 26 adults aged 20–28 years (19 males and 7 females) with varied heights and weights. As shown in Figure 11b, each participant slept for approximately three hours in a dark, quiet room designed to resemble a typical bedroom. Before the session, we performed participant-specific perceptual calibration at 20 Hz, selected based on structural characterization.

Reference measurements were collected using a Muse headband [4, 7, 42], a respiration belt [20], and, with consent, a camera for monitoring motion and posture. We compare sleep-stage estimates against mmWave, Fitbit Charge 6 [1], and Google Nest Hub [31]. Both commercial devices provide a sleep stage every 30s. Six sessions were excluded because of acquisition failures, including invalid Fitbit data and mmWave recordings, leaving 20 participants (13 male and 7 female) for analysis.

*5.2.1 Epoch-Level KFold Validation.* The partial-night dataset contains 6,990 labeled 30s epochs for *SleepVibe* and 7,279 for the mmWave baseline, which is 33.5% awake, 11.9% REM, 43.5% light, and 11.1% deep. We follow the shuffled, stratified 10-fold procedure introduced in Section 5.1.1, using eight folds for training, one for validation,

and one for testing. Across sequence lengths and folds, this yields 5,120–5,824 epochs for training and 640–728 epochs for each validation and test partition. Sequences are non-overlapping with train-fold normalization, and the best validation checkpoint is restored before testing. We repeat the evaluation using five random seeds. We configure the focal loss with $\gamma = 2$ and inverse-frequency class weighting with $\rho = 1$ for the partial-night study.

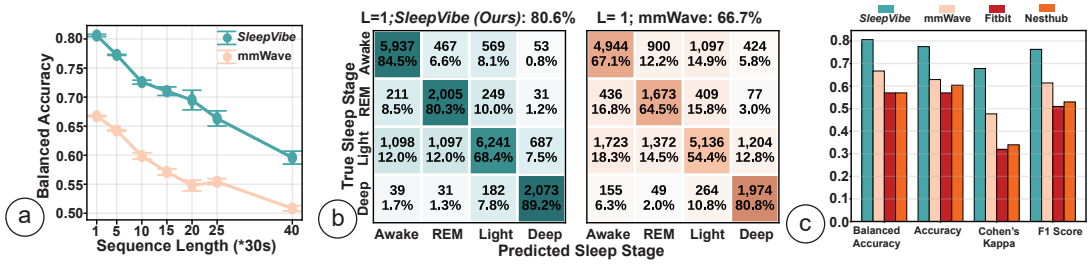



Fig. 15. Partial-night KFold results: (a) sequence-length performance, (b) confusion matrices at $L = 1$, and (c) comparison with all baselines.

*SleepVibe* outperforms mmWave at every sequence length (Figure 15a). Its balanced accuracy varies from 80.6% at $L = 1$ to 59.6% at $L = 40$, remaining 13.9 and 8.8 percentage points above mmWave, respectively. Over the same range, the fold-level standard deviation of macro-F1 rises from 1.1 to 9.9 percentage points for *SleepVibe* and from 1.9 to 10.8 percentage points for mmWave.

At $L = 1$, *SleepVibe* provides higher recall for all four sleep stages (Figure 15b). Light has the lowest recall for both modalities because the imbalance-mitigation objective deliberately redistributes performance toward minority stages, as discussed in Section 5.3. As shown in Figure 15c, *SleepVibe* also achieves Cohen's $\kappa = 0.678$ and an F1 score of 76.3%, versus 0.477 and 61.4% for mmWave. Compared with common off-the-shelf sleep-tracking devices, *SleepVibe* achieves a balanced accuracy of 80.6%, compared with 57% for Nest Hub and Fitbit Charge 6.

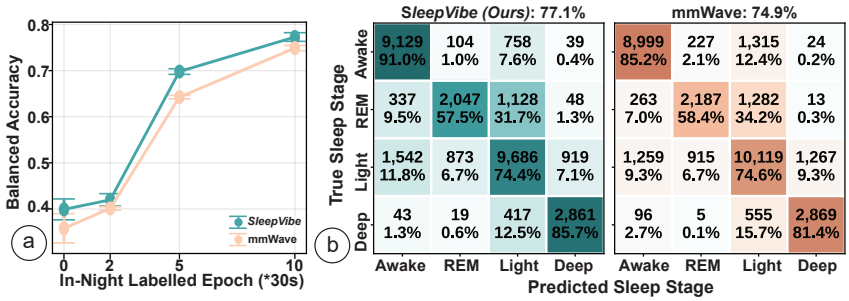



Fig. 16. Partial-night LOSO at $L = 40$: (a) performance with up to 10 calibration epochs, and (b) confusion matrices with 10 epochs.

*5.2.2 Leave-One-Subject-Out Evaluation.* We next assess generalization to an unseen participant in the broader partial-night cohort with and without labeled participant-specific calibration. In each of the 20 LOSO folds, one participant is held out from initial training and validation, two are used for validation, and the remaining 17 are used for training. We evaluate under $L = 40$ and repeat the entire procedure using five random seeds.

For calibration, we select 2, 5, or 10 labeled 30 s epochs at evenly spaced positions across the held-out recording. The selection depends only on recording length and is independent of sleep-stage labels and random seeds. These calibration epochs are excluded from evaluation, ensuring that all calibration budgets, including the uncalibrated

baseline, are evaluated on the same test epochs. We then fine-tune a copy of the LOSO model end-to-end for 30 full-batch AdamW steps using the same focal and class-weighted objective as in training, with a learning rate of $3 \times 10^{-4}$ and weight decay of $10^{-4}$. Under the $L = 40$ configuration, only the selected calibration epochs contribute to the loss, while labels for all other epochs are masked. To ensure a consistent comparison across calibration budgets, we use a fixed number of optimization steps without early stopping.

Without participant-specific calibration, balanced accuracy remains around 40% for both modalities (Figure 16a), using no labels from the held-out participant for model training. This limited performance may be partly due to the limited coverage of each participant's sleep-stage progression and the absence of night-to-night variation, making subject-independent transfer particularly difficult. To assess whether sparse participant-specific calibration can improve this transfer, we use 10 labeled epochs, corresponding to five minutes of labeled recording distributed across the session. Balanced accuracy increases to 77.1% for *SleepVibe* and 74.9% for mmWave (Figure 16b). This result shows that sparse calibration can recover much of the participant-specific information missing from a single partial-night recording.

## 5.3 Impact of Data Imbalance

Table 3. Impact of loss configuration on sleep-stage classification performance.

| **Loss** | **Per-stage Recall (Full)** | | | | **Bal. Acc.** | **Per-stage Recall (Partial)** | | | | **Bal. Acc.** |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Awake** | **REM** | **Light** | **Deep** | | **Awake** | **REM** | **Light** | **Deep** | |
| Stage proportion | 7.4% | 26.1% | 46.0% | 20.5% | – | 33.5% | 11.9% | 43.5% | 11.1% | – |
| Plain CE | .676±.004 | .794±.004 | .801±.007 | .816±.004 | .772±.003 | .858±.003 | .636±.018 | .821±.005 | .792±.002 | .777±.004 |
| Inv.-freq. + focal | .690±.005 | .798±.004 | .797±.005 | .821±.006 | **.776±.001** | .845±.003 | .803±.008 | .684±.004 | .892±.005 | **.806±.002** |

Due to the natural sleep pattern, both studies are dominated by Light sleep, but their minority classes differ: Awake is rarest in the full-night study, whereas REM and Deep are rarest in the partial-night study (Table 3). We evaluate the effect of class imbalance at $L = 1$ so that the comparison is not confounded by cross-epoch temporal context. Our mitigation strategy retains every training label while combining focal loss with inverse-frequency class weighting. The full-night model uses $\gamma = 0.1$ and $\rho = 0.05$, whereas the partial-night model uses $\gamma = 2$ and $\rho = 1$ to place greater emphasis on underrepresented REM and Deep epochs. The benefit is most pronounced in the partial-night setting: balanced accuracy increases from 77.7% to 80.6%, with gains of 16.7 and 9.9 percentage points in REM and Deep recall, respectively. In the full-night study, balanced accuracy remains slightly increased from 77.2% to 77.6%. The accompanying reduction in Light recall indicates that the class-aware objective redistributes performance toward the underrepresented stages, maintaining or improving overall balanced accuracy.

## 5.4 Impact of Feature Selection

To characterize the impact of physiological feature representation on sleep stage classification, we evaluate *SleepVibe* against reduced feature sets relying on rate-level metrics and coarse movement cues. We compare the complete representation with three reduced feature sets: (1) heart rate, computed from the dominant frequency of the two heartbeat channels; (2) respiratory rate, computed analogously from the respiration channels; and (3) movement features comprising body-movement magnitude, combined movement and heart-rate changes, and motion-channel standard deviation. We use $L = 1$ to isolate the contribution of the feature representation from temporal context. Figure 17 indicates that each reduced representation achieves an accuracy near or below 60%, similar to Fitbit and Nest Hub in Figure 15. These results indicate that rate-level summaries and coarse motion

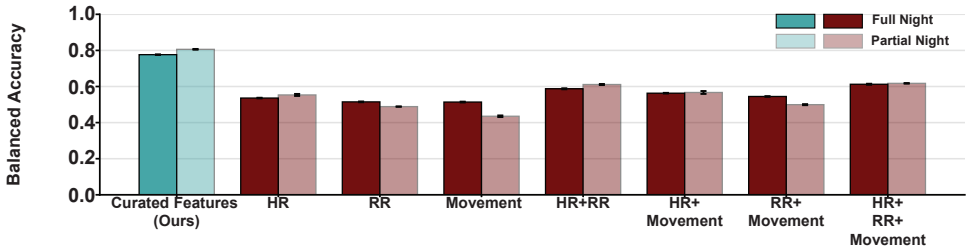



Fig. 17. Sleep-stage classification using *SleepVibe*'s curated representation, rate-level features, or coarse movement cues.

cues alone remain limited for reliable sleep stage monitoring. Instead, the performance gains of *SleepVibe* arise from its ability to leverage actively probed vibration responses to extract physiologically meaningful features at the waveform level.

## 5.5 Respiration of Real Sleep Session

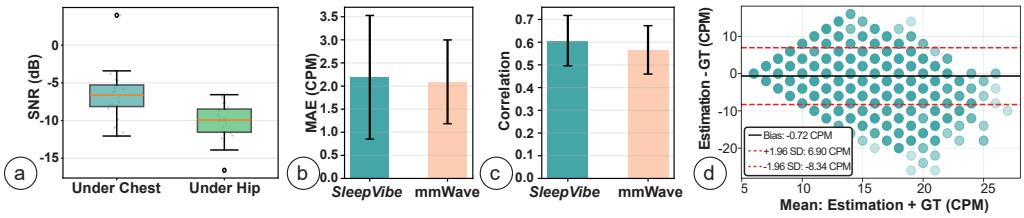



Fig. 18. Respiratory effort evaluation across real sleep sessions. (a) The geophone under the chest provides the higher respiratory SNR on average. (b)&(c) *SleepVibe* achieves respiratory rate accuracy and waveform-level correlation comparable to mmWave sensing. (d) Bland-Altman analysis of *SleepVibe* respiratory rate estimation.

The partial-night recordings allow us to evaluate respiratory sensing during natural sleep at both the rate and waveform levels. We analyze 30s windows and estimate respiratory rate from the dominant FFT peak between 0.1 and 0.4 Hz. To separate physiological monitoring from gross motion, we exclude windows containing more than 10s of movement, which accounts for 1.2% of the dataset. Within each remaining window, we use the sensing channel with the higher SNR, defined as respiration-band energy divided by out-of-band energy. The chest channel provides the higher participant-averaged SNR (Figure 18a), consistent with stronger respiration-induced pressure changes near the chest.

As shown in Figure 18b, using this channel-selection procedure, *SleepVibe* achieves a respiratory rate error of 2.19 cpm, comparable to 2.09 cpm for mmWave. Its waveform correlation is 0.61, compared with 0.57 for mmWave, as shown in Figure 18c. Figure 18d presents a Bland–Altman analysis between *SleepVibe* and the ground-truth respiration signal, illustrating the agreement and bias distribution across all evaluated epochs. Overall, 93.4% of *SleepVibe* respiratory rate estimates fall within the Bland-Altman limits of agreement, demonstrating a strong alignment between *SleepVibe* and ground truth prediction. These results suggest that active vibration sensing enables stable recovery of respiration waveforms during real sleep sessions. Importantly, participants naturally transitioned between multiple sleeping postures, indicating robustness across different postures, as further evaluated in Section 5.6.1.

## 5.6 Characterization of Respiration and Heartbeat Activity

We conduct a controlled session in which participants remain stationary in predefined postures and alternate between light and deep breathing. The fingertip PPG signal provides the reference heartbeat waveform for this evaluation. We select the segment with the higher SNR and identify its dominant FFT peak within 0.7–2.0 Hz [8] as the heart rate estimate. The mmWave is also employed as the contactless baseline [40]. For each bed structure, we select an excitation frequency based on the structural characterization and keep it fixed throughout the corresponding robustness evaluation.

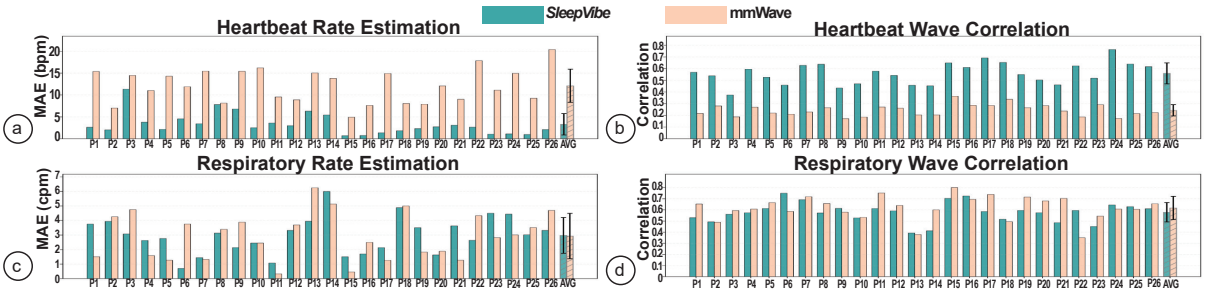



Fig. 19. Controlled physiological evaluation across 26 participants. Panels (a) and (b) show the heart rate MAE and waveform correlation, respectively. Panels (c) and (d) show the respiratory rate MAE and waveform correlation. *SleepVibe* significantly outperforms mmWave in heartbeat monitoring (two-sided $t$-test, $p < 0.001$) while achieving comparable respiratory monitoring performance.

As shown in Figure 19a, across 26 participants, *SleepVibe* achieves a heart-rate MAE of 3.3bpm, compared with 12.08 bpm for mmWave. This error is below the 4.7 bpm reported for common consumer wearables [49] and satisfies the ±5 bpm or ±10% criterion for cardiac monitors [26]. Respiratory performance is comparable across modalities: *SleepVibe* and mmWave achieve MAEs of 2.97 and 2.93 cpm, respectively (Figure 19c), both within the reported ±3.9 cpm tolerance [50]. The waveform correlation results lead to the same conclusion. After linear scaling, *SleepVibe* achieves Pearson correlations of 0.56 for heartbeat and 0.58 for respiration, compared with 0.24 and 0.62 for mmWave, as shown in Figure 19b&d. Thus, *SleepVibe* preserves substantially more cardiac waveform structure while providing similar respiratory fidelity.

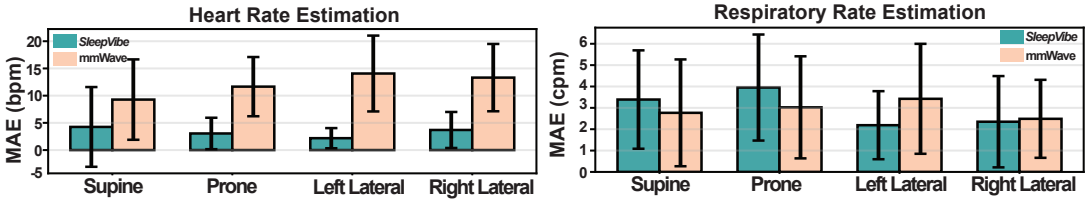



Fig. 20. Effect of sleep posture on heartbeat and respiratory-rate estimation.

*5.6.1 Impact of Different Sleep Postures.* We evaluate the robustness of *SleepVibe* across different sleep postures to understand how body orientation affects the reconstruction of physiological processes. Figure 20 summarizes the impact of posture on heartbeat and respiratory rate estimation across four representative positions: supine, prone, left lateral, and right lateral.

Across all postures, *SleepVibe* consistently outperforms mmWave in heartbeat rate estimation, indicating that *SleepVibe* reliably extracts the heartbeat activity coupled to the bed frame regardless of body orientation. For

respiration, the relative performance between the two systems depends on posture. While *SleepVibe* slightly underperforms mmWave in the supine and prone postures, it surpasses mmWave when participants lie on their left or right side. This contrast reflects the different sensing mechanisms: mmWave is primarily sensitive to chest displacement along its line of sight, which is most pronounced when the participant faces upward or downward. When the participant lies laterally, thoracic motion becomes largely orthogonal to the radar, leading to degraded mmWave performance. In contrast, *SleepVibe* relies on active vibration sensing to probe respiration-induced vibration responses, which remain observable through the bed structure even when thoracic motion is not directly facing the sensor. The partial-night study includes a wider range of unconstrained sleep postures, as discussed in Appendix E. Although we do not conduct a separate posture-stratified analysis of these natural recordings, *SleepVibe* maintains the highest classification performance among the evaluated systems, suggesting resilience to the posture variability encountered during real sleep.

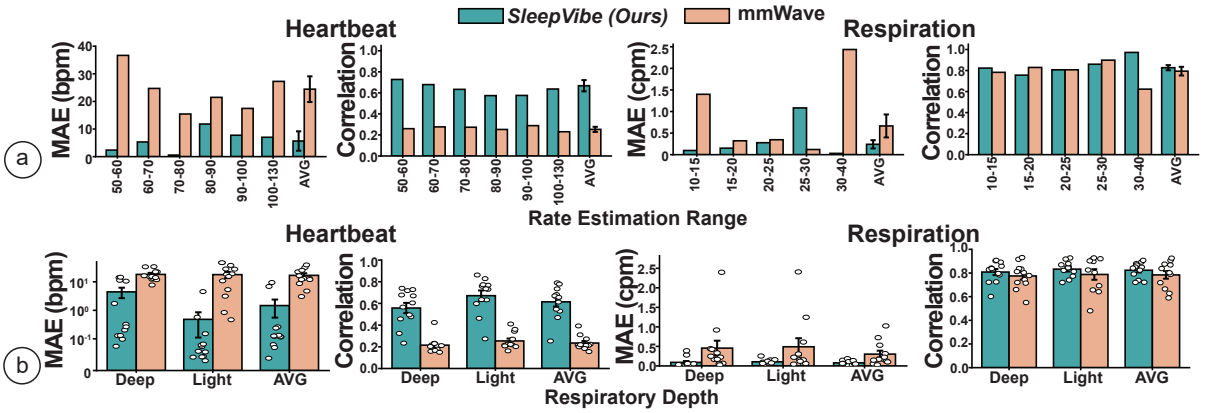



Fig. 21. Robustness to physiological variation. (a) Elevated heart and respiratory rates. (b) Deep and light breathing.

*5.6.2 Impact of Higher Heart and Respiratory Rates.* To test whether performance extends to elevated physiological rates, we evaluate 12 additional participants with a post-exercise recovery study. Each participant is asked to perform a 30s high-kick and is monitored in a supine position for 120s during recovery. This results in respiratory rates of 7.1–36.8 cpm and heart rates of 50.5–120.2 bpm. To accommodate these ranges, respiration and heartbeat use filters and dominant-frequency search bands of 0.1–0.7 Hz and 0.7–3 Hz, respectively.

As Figure 21a suggests, *SleepVibe* remains robust and more accurate than mmWave at these elevated rates. Across the full heart-rate range, *SleepVibe* achieves an MAE of 5.66 bpm and a waveform correlation of 0.67, which is about a 4.5x improvement compared with mmWave (24.44 bpm and 0.25). The error remains comparable to the ±5 bpm or ±10% criterion [26]. Within the 100–130 bpm range, *SleepVibe* maintains an MAE of 7.02 bpm and a correlation of 0.64, whereas mmWave reaches 27.26 bpm and 0.23.

The respiratory results are similarly stable. *SleepVibe* achieves a respiratory-rate MAE of 0.24 cpm and a waveform correlation of 0.83, compared with 0.67 cpm and 0.79 for mmWave, and remains below the reported ±3.9 cpm tolerance [50]. One nonstationary window in the 25–30 cpm range contains competing spectral peaks: the belt selects 25.35 cpm, whereas *SleepVibe* selects 20.38 cpm but retains a secondary peak at 25.20 cpm with 91% of the selected peak's magnitude and a waveform correlation of 0.90. At the highest observed respiratory-rate interval of 30–40 cpm, *SleepVibe* further achieves a 0.03 cpm MAE and a 0.97 waveform correlation.

*5.6.3 Impact of Respiratory Depth.* *SleepVibe* preserves accurate physiological sensing under both deep and light breathing and outperforms the mmWave baseline across all four metrics shown in Figure 21b. We evaluate

12 participants during two 60s deep breathing trials and two 60s light breathing trials in the supine posture. *SleepVibe*'s MAEs of respiratory rate estimation remain 0.09 and 0.10 cpm, compared with 0.46 and 0.49 cpm for mmWave, which is about a 5x improvement. It also achieves respiration waveform correlations of 0.81 and 0.83, slightly exceeding the corresponding mmWave correlations of 0.78 and 0.79.

Deep breathing has a larger impact on heartbeat sensing because stronger respiration modulates the amplitude envelope of heartbeat-induced vibrations [35]. Consequently, *SleepVibe*'s heartbeat waveform correlation decreases from 0.67 under light breathing to 0.56 under deep breathing, but remains well above mmWave at 0.25 and 0.22. *SleepVibe* also maintains lower heart rate MAE under both deep breathing (3.34 versus 18.21 bpm) and light breathing (0.48 versus 17.88 bpm). Together with the post-exercise evaluation in Section 5.6.2, they demonstrate robust heartbeat and respiration sensing across variations in physiological rate and respiratory depth.

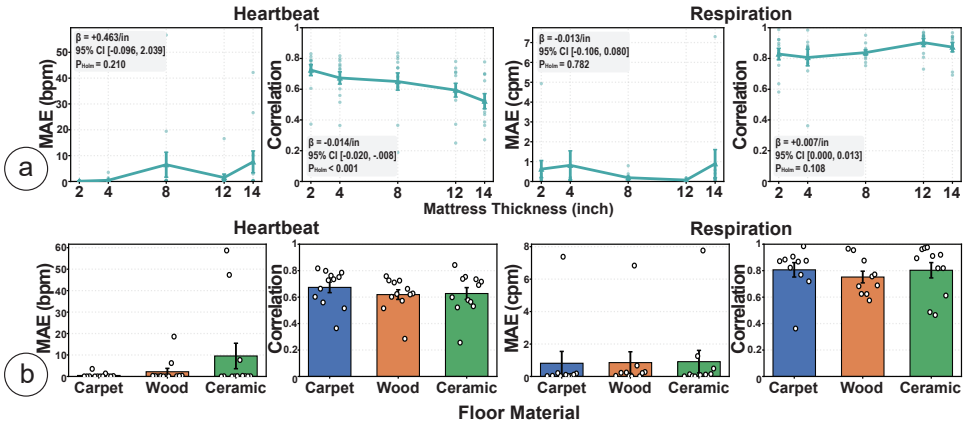



Fig. 22. Robustness across (a) mattress thicknesses and (b) floor materials.

*5.6.4 Impact of Mattress Thickness.* As mattress thickness can attenuate structure-borne responses, we test 12 participants on mattresses ranging from 2 to 14 inches. Each participant performs one 30s supine, normal-breathing recording. For heart rate MAE, respiratory rate MAE, and the heartbeat and respiration waveform correlations, we fit linear mixed-effects models [27] with mattress thickness as a continuous fixed effect and participant-specific random intercepts. We apply Holm correction across the four tests [2].

Mattress thickness does not significantly affect respiratory rate MAE ($p_{\text{Holm}} = 0.782$) or respiration correlation ($p_{\text{Holm}} = 0.108$). Even at 14 inches, *SleepVibe* achieves an MAE of 0.89 cpm and a correlation of 0.87, as shown in Figure 22. Heartbeat correlation declines from 0.72 at 2 inches to 0.52 at 14 inches, yielding a significant thickness effect ($p_{\text{Holm}} < 0.001$), due to attenuation of weaker cardiac vibrations in thicker mattresses. However, thickness does not significantly affect heart rate MAE ($p_{\text{Holm}} = 0.210$), which is 3.16 bpm on average across conditions. Thus, thicker mattresses reduce fine-grained heartbeat morphology, yet *SleepVibe* continues to provide accurate heart rate estimates and highly reliable respiration sensing even under a 14-inch mattress.

*5.6.5 Impact of Floor Material.* Floor material changes the support boundary and damping of the bed structure. We therefore evaluate 12 participants on three common floor materials: carpet, wood, and ceramic, with one 30s recording per condition. As shown in Figure 22b, respiration sensing remains highly consistent across the three materials: mean respiratory rate MAE ranges only from 0.81 to 0.91 cpm, while mean respiration waveform correlation ranges from 0.75 to 0.81. Heartbeat waveform correlation is also stable between 0.62 and 0.67. The corresponding mean heart rate MAEs are 0.46, 2.20, and 9.55 bpm on carpet, wood, and ceramic, respectively. The higher error on the ceramic floor is dominated by two isolated peak-selection errors of 58.69 and 47.32 bpm.

Yet, its median error is 0.11 bpm, comparable to 0.06 and 0.20 bpm on carpet and wood. Using floor material as a categorical fixed effect and participant as a random intercept, the linear mixed-effects models find no significant material effect on any of the four outcomes after Holm correction (all $p_{\text{Holm}} \geq 0.593$). These results demonstrate that *SleepVibe* maintains robust physiological sensing across common flooring surfaces despite their different structural damping properties.

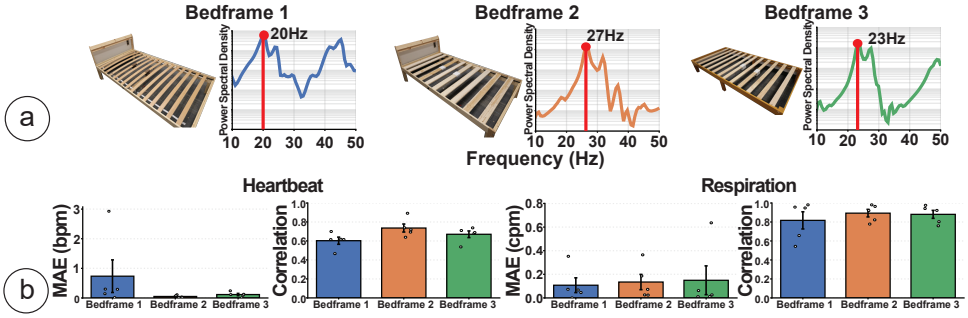



Fig. 23. Evaluation across bed frames. (a) Tested bed frames and excitation frequencies selected through impulse-response characterization. (b) Heartbeat and respiratory performance.

*5.6.6 Impact of Bed Frame.* As bed frames exhibit different resonant responses, practical deployment requires a structural characterization procedure that can be applied across bed-frame configurations to select an appropriate excitation frequency. As shown in Figure 23a, we evaluate five additional participants lying in the supine posture and breathing normally on three different bed frames. For each frame, we analyze the impulse response of the instrumented beam and select its dominant resonant frequency as the excitation carrier. This procedure identifies excitation frequencies of 20 Hz for bed frame 1, 27 Hz for bed frame 2, and 23 Hz for bed frame 3, as shown in Figure 23a. Importantly, none of the participants reported perceiving the active excitation after calibration. Figure 23b shows that *SleepVibe* maintains consistent respiratory sensing performance across the three configurations, with mean respiratory rate MAEs of 0.11, 0.14, and 0.15 cpm and mean respiration waveform correlations of 0.82, 0.89, and 0.88, respectively. The corresponding mean heartbeat waveform correlations are 0.603, 0.736, and 0.671, while the mean heart rate MAEs are 0.74, 0.05, and 0.12 bpm. The higher mean heart rate MAE for Bed Frame 1 is driven by a single peak-selection error of 2.93 bpm, and its median MAE remains comparable to those of bed frames 2 and 3 at 0.29 bpm. Linear mixed-effects analysis reveals no significant differences in heart rate MAE, respiratory rate MAE, or respiration waveform correlation after Holm correction. The only significant pairwise contrast is a higher heartbeat waveform correlation for bed frame 2 than for bed frame 1 ($p_{\text{Holm}} < 0.001$). These results indicate that *SleepVibe* generalizes across the tested bed frames through a brief frame-specific structural characterization, rather than requiring a fixed bed structure or excitation frequency.

## 6 Discussion and Future Work

### 6.1 Scope of the Sleep-Stage Reference

Our evaluation uses sleep-stage labels generated by a Muse S headband [4, 7, 42] rather than labels obtained from simultaneous clinical PSG. Recent studies indicate that Muse S sleep staging performance achieves substantial agreement with PSG results, with 0.76–0.79 Cohen's kappa and about 88% accuracy [4, 42]. We position the current results as performance relative to a consumer-grade EEG reference and as a controlled comparison with the wearable and contact-free baselines evaluated under the same sessions, rather than as evidence of clinical or diagnostic accuracy. Future work will evaluate *SleepVibe* using simultaneously collected PSG labels [29], including participants with sleep-related disorders.

### 6.2 Heart Rate Estimation and Scope of Robustness Evaluation

Ambiguous spectral peak selection can produce occasional large heart rate errors that disproportionately increase the mean error. Across the physiological and environmental robustness evaluations, 5.21% windows produced estimates above 1.5 times or below two-thirds of the PPG reference rate. This is due to spectral ambiguity between the cardiac fundamental and its harmonics, or other competing peaks. For example, in one ceramic-floor recording, the estimated heart rate is 120.98 bpm, approximately twice the PPG reference of 61.29 bpm, although a competing peak remains visible at 60.78 bpm. During longer monitoring, temporal consistency across successive windows could help reject isolated rate jumps, while agreement across sensors could help distinguish competing peaks.

For sleep stage monitoring, complementary physiological features and temporal context may reduce the classifier's sensitivity to such isolated rate errors, but this robustness remains to be validated. Our model combines cardiac and respiratory waveform characteristics, variability, and movement features with information from surrounding epochs. However, the controlled evaluations of physiological variations, mattress thicknesses, floor materials, and bed frames assess signal reconstruction and rate estimation in awake participants lying supine. They therefore do not establish classification robustness across these conditions, where sleep posture, physiological state, and environmental factors may interact. Future work will quantify the downstream effects of rate errors and evaluate sleep-stage classification across these physiological ranges and deployment conditions.

### 6.3 Generalization to Older Adults and Individuals with Sleep-Related Disorders

Our sleep-stage evaluation is limited to young adults without known severe sleep-related disorders. Age and health status can alter sleep architecture, cardiorespiratory dynamics, movement, body–bed coupling, and sensitivity to active excitation [10, 47, 55]. These population-dependent differences may shift the vibration features observed by *SleepVibe* and reduce classifier performance without adaptation. As heartbeat, respiration, and movement still mechanically couple into the bed, the underlying sensing principle still remains applicable [47]. Future work will evaluate these populations and adapt excitation signal characteristics (e.g., frequency, amplitude) and modeling accordingly. For example, *SleepVibe* could select among multiple candidate bed-frame resonances to preserve respiration sensitivity and heartbeat separation while remaining imperceptible across age-dependent perceptual thresholds [28]. For individuals with sleep apnea, respiratory pauses, irregular effort, and abrupt cardiac responses, the current sleep stage classification may be confused. Supporting these cases will require apnea-aware features, processing for nonstationary breathing, and population-specific training.

### 6.4 Generalization to Bed Frames with Various Materials

Our evaluation covers multiple wooden bed frames, while future work should examine other materials, such as steel. Compared with wide wooden supports, narrower steel beams may reduce body–structure coupling and attenuate physiological vibrations at a single sensing location. To improve compatibility across frame geometries, we envision an extended bracket spanning two adjacent beams, with a sound exciter mounted underneath and an L-shaped sensing arm suspended in the gap to support the geophone. This design could increase the effective coupling area and collect heartbeat vibrations through multiple structural paths, benefiting narrow frames or thick mattresses. As spatial phase differences may also cause destructive averaging, future work will characterize the bracket geometry and mounting locations through bed-specific frequency-response measurements and compare its signal gain with the current single-beam design.

### 6.5 Multi-User Sleep Stage Monitoring

We aim to extend *SleepVibe* to scenarios where the bed is shared by two or more people, such as couples or families. In these cases, physiological vibrations from different users become naturally mixed on the structure, making it challenging to isolate individual signals. Prior work has explored blind source separation for multi-user

sleep monitoring [35], showing the feasibility of disentangling overlapping signals. Building on this insight, we would like to explore leveraging multiple geophone sensors and adapting techniques from microphone array processing, such as beamforming, to target specific users and extract individual vital signals for sleep stage classification. By distinguishing each individual's sleep stage, we can move beyond single-user analysis to also study how sleep patterns interact with each other and model the dynamics of co-sleeping, such as whether disturbances from one person's movements or sleep stage transitions affect the other's rest.

### 6.6 Sleep Intervention Therapy with Sleep Monitoring

Beyond sensing, *SleepVibe* can be extended to support active sleep intervention therapies. Our system already incorporates an active interrogation vibration signal, paving the way for using vibration stimuli as a form of closed-loop feedback to enhance sleep quality. For example, recent studies have shown that subtle mechanical cues delivered during specific sleep stages are effective in alleviating sleep inertia and facilitating cognitive recovery following sleep [68]. By integrating real-time sleep stage monitoring with vibration-based interventions embedded in bedding fabrics [46, 48] or structures [88], we envision a system that not only tracks but also improves sleep quality, which transforms *SleepVibe* from a sensing system into a sleep health platform.

### 6.7 Ethical Considerations

*SleepVibe* reduces privacy exposure by sensing structure-borne vibrations rather than recording images or audio. Yet, the inferred sleep stages, physiological activity, movement, and occupancy remain sensitive health and behavioral data. Future deployments should therefore prioritize local processing, minimize raw-signal retention, encrypt stored and transmitted data, and provide clear user control over sensing and data access. Shared-bed use should also require consent from all occupants, visibly indicate when sensing is active, and provide accessible pause and power-off controls, with individual-level outputs restricted to each consenting user. In addition, active structural probing should remain within the selected frequency range and below each sleeper's perceptual threshold through authenticated actuator control, hardware-enforced amplitude limits, and a physical disable option. Characterizing vibration propagation will further help define the effective sensing region and support privacy-aware deployment.

## 7 Conclusion

We presented *SleepVibe*, an end-to-end sleep stage monitoring system that uses active structure-borne vibration sensing to recover heartbeat, respiration, and body movement across diverse sleeping postures. Through structure-guided active probing and a physiology-aware feature representation, *SleepVibe* separates weak physiological responses and supports sleep-stage inference beyond rate-level summaries.

Evaluated against a consumer-grade EEG headband across 34 young adults aged 20–28 with no reported severe sleep-related disorders, *SleepVibe* achieves 77.6% and 69.9% balanced accuracy under epoch-level KFold and LOSO evaluation under the full-night study, outperforming mmWave by 10.9 and 6.4 percentage points, respectively. It also achieves higher Deep sleep recall (82.1% and 72.0%) under both settings against mmWave (71.3% and 60.7%). Beyond sleep stage monitoring, *SleepVibe* estimates heart rate with an MAE of 3.3 bpm, approximately a 4x improvement over mmWave, while achieving comparable respiratory rate accuracy (2.97 cpm). The physiological signal reconstruction remains robust to elevated physiological rates, different wooden bed frames and floor materials, and mattresses up to 14 inches thick. These results demonstrate the potential of active structural vibration sensing as a privacy-preserving, comfortable, and scalable approach to long-term sleep monitoring.

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## A Physiological Feature Definitions

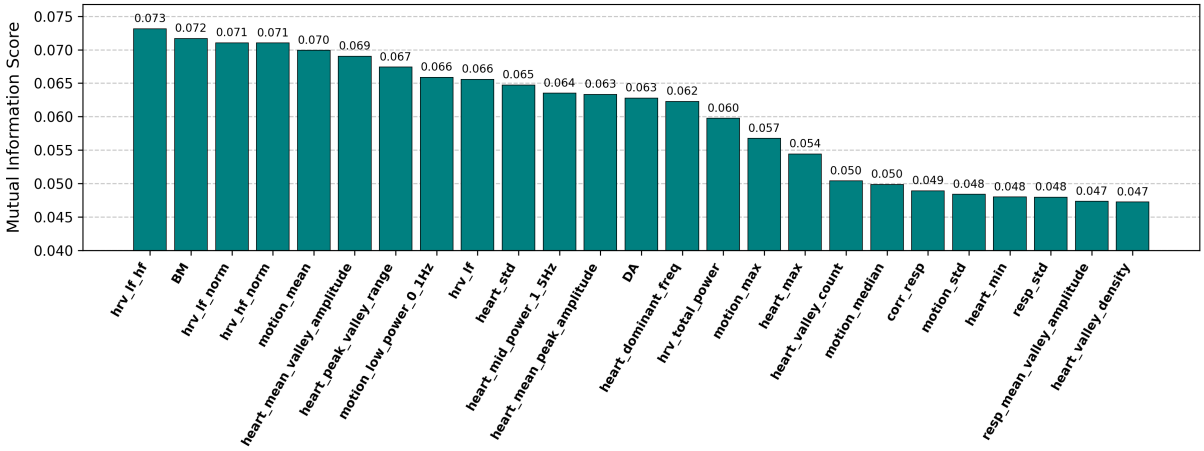



Fig. 24. Top 25 physiological features aggregated across sensing channels.

We characterize how the representation relates to sleep stage by computing the mutual information [39] between each feature and the sleep-stage labels and averaging the scores of the same feature type across sensing channels. Figure 24 presents the 25 highest-scoring feature types. The ranking shows that sleep-stage information is distributed across heartbeat, respiration, and movement cues. Heartbeat rhythm and morphology features, especially HRV [73], appear prominently, while respiratory cross-channel correlation and morphology features also carry sleep-related information. Movement statistics and physiologically motivated measures, such as body movement (BM), provide an additional complementary cue. Together, these results support a multi-process representation that captures more than physiological rates alone. The complete set of 83 standardized features is provided to the MLP–BiGRU, which learns nonlinear feature interactions and temporal dependencies across consecutive windows.

## B Validation of Mechanical Amplification

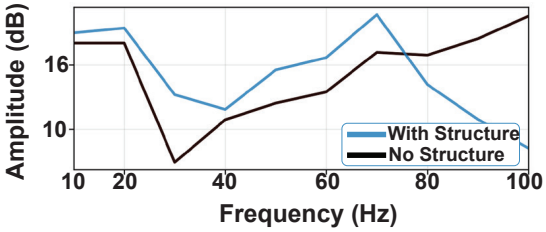



Fig. 25. The amplification effect of the C-shaped frame.

We validate the amplification effect of the C-shaped frame supporting the geophone sensor by using a frequency sweep signal. We compare vibration amplitudes measured by a sensor placed directly on the bed frame and one mounted on the C-shaped structure. A frequency sweep from 0–240 Hz is transmitted using the sound exciter.

As shown in Figure 25, the sensor mounted on the C-shaped structure exhibits higher vibration amplitude than the bed-mounted sensor over a broad frequency range, with pronounced amplification between 10 Hz and 70 Hz. This range covers the dominant frequency bands of the heartbeat-induced vibration and excitation signals identified in the structural characterization.

## C Impact of Height

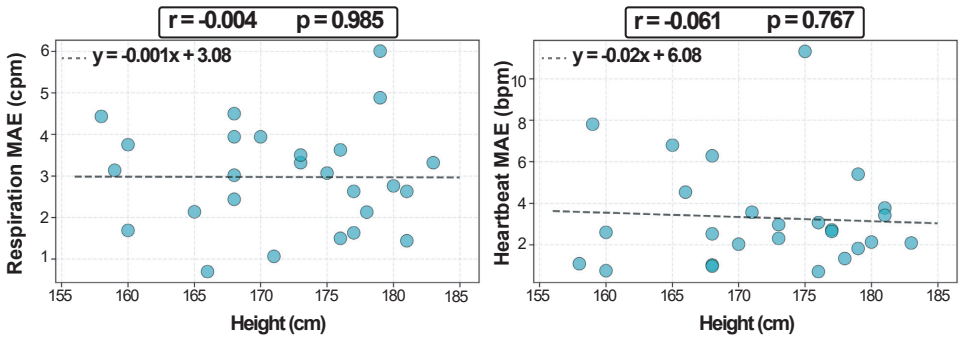



Fig. 26. Correlation between participant height and physiological signal reconstruction accuracy.

Sensor placement is guided by the approximate chest and hip regions, where sleep-related physiological processes couple most strongly to the bed frame. Thus, we evaluate the impact of participant height on *SleepVibe* using a fixed hardware configuration. Variations in body height can shift the relative positions of the chest and hip with respect to the mounted sensors, potentially affecting signal coupling. To assess robustness to this variation, we analyze the relationship between participant height and the accuracy of heartbeat and respiratory rate estimation. As shown in Figure 26, correlation analysis reveals no statistically significant association between participant height and estimation accuracy. In both analyses, the correlation coefficients are close to zero, and the corresponding p-values are significantly greater than 0.05, indicating the absence of a meaningful linear relationship. These results suggest that *SleepVibe* maintains robust performance across a range of participant heights, despite fixed sensor placement.

## D Impact of Channel Selection

We analyze the impact of sensing channel selection on sleep stage classification to understand the trade-off between sensor deployment cost and classification performance. Starting from the current 83-feature representation, each single-channel configuration retains the 53 nonredundant features that can be computed from the selected vibration channel and motion signal, excluding features that require both vibration channels. Under the $L = 1$ KFold setting, the chest- and hip-mounted channels achieve 75.6% and 73.5% accuracy in the full-night study, resulting in a performance drop of approximately 2–3%. In the partial-night study, they achieve 70.3% and 69.5%, compared with 77.5% using both channels. Despite the reduction, both single-channel configurations continue to outperform the mmWave baseline in both studies. These results indicate that rich sleep-related information is locally encoded in each vibration channel, enabling a simpler sensing configuration, while combining both channels provides the highest overall accuracy.

## E Posture Changes in Sleep Session

Various postures captured during natural partial-night sleep.

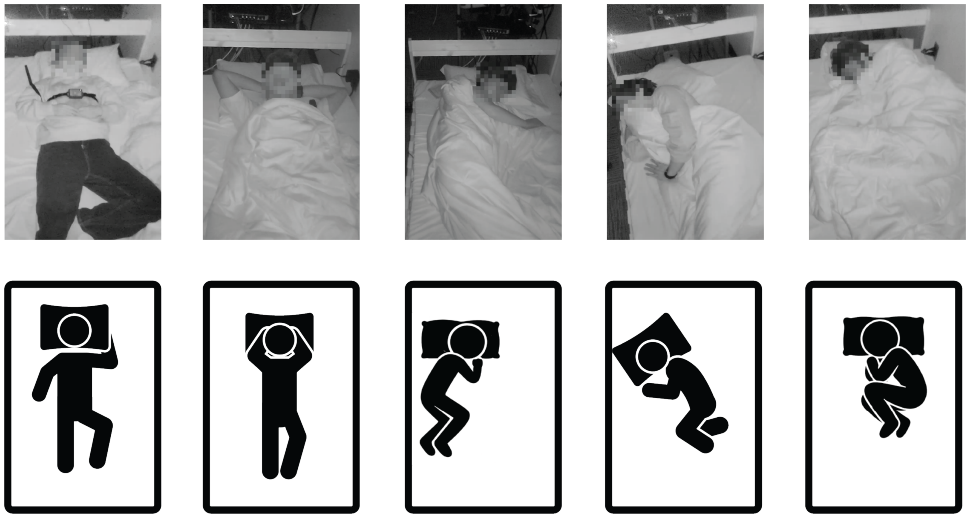


Fig. 27. Different sleep postures under real sleep sessions.

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